
コンピュータビジョン特論A/B
現代脳計算論

Computer Vision A/B
Modern Neural Computation

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Local Feature/Corner Detector

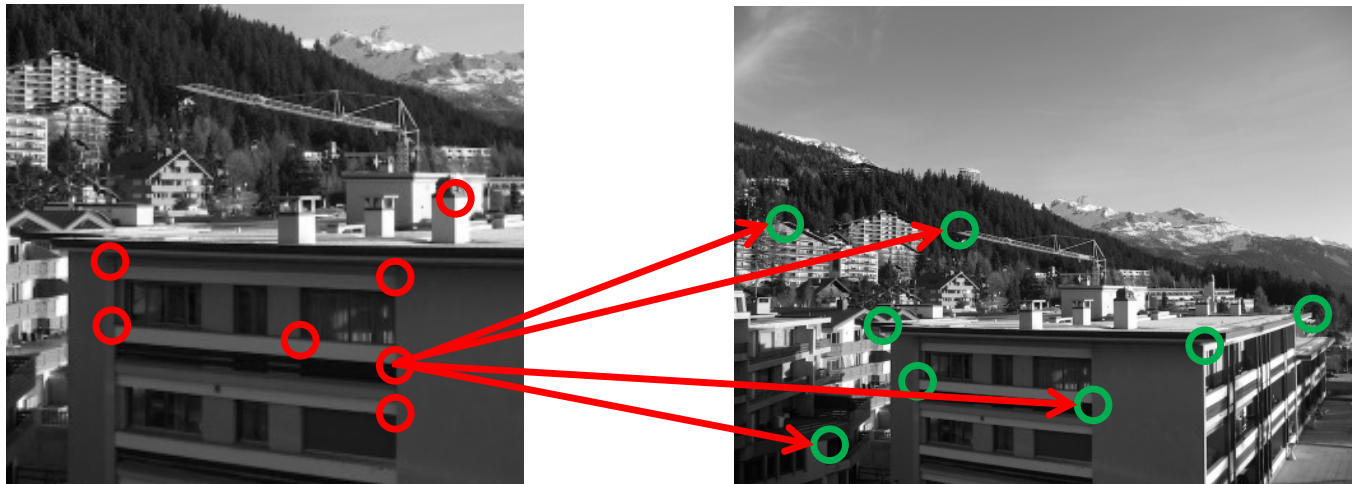
54

- ❑ Local Feature & Feature Detector
 - SIFT/SURF Feature (brief explanation)
- ❑ Find Corresponding Key points
 - Image matching
 - Template matching
- ❑ Other image features
 - HoG (Histogram of Oriented Gradients)
 - Haar-like
 - ◆ Face Detection

Feature Detection/Feature Description

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Local Feature Matching



Which feature point is the corresponded point?

Feature Description

- Feature vector should be represented as a unique vector
 - ◆ Rotation, Scale, illumination, Affine transformation, ...

Invariant Feature

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□ Harris Corner, FAST Corner

- Rotation Invariant Feature

□ SIFT: Scale-Invariant Feature Transform [D. Lowe, 1999, 2004]

- Rotation and Scale Invariant
- Robust for noise and Illumination change
- 128 feature vector for each key point
 - ◆ Normalized histogram of gradient direction

□ SURF: Speeded Up Robust Feature [H. Bay, T. Tuytelaars, L. V. Gool, 2006]

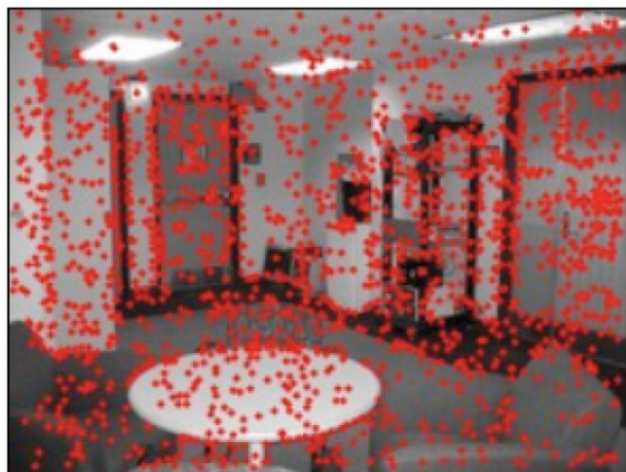
- Fast rotation and scale invariant feature
- Almost 10 times faster than SIFT

SIFT (1/4)

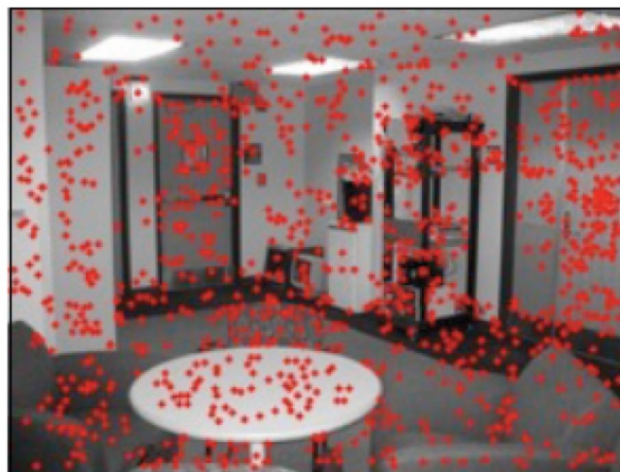
57

□ SIFT: Scale-Invariant Feature Transform [D. Lowe, 1999, 2004]

- Rotation and Scale Invariant
- Robust for noise and Illumination change
- 128 feature vector for each key point
 - ◆ Normalized histogram of gradient direction



キーポイント候補点



主曲率による候補点削除



コントラストによる候補点削除

[D. Lowe,, 2004]

[藤吉, 第76回ロボット工学セミナー, 2013]

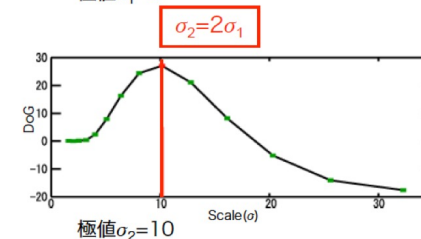
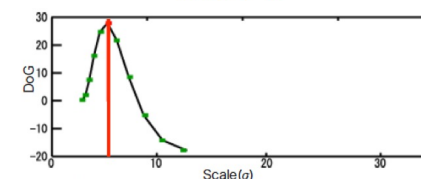
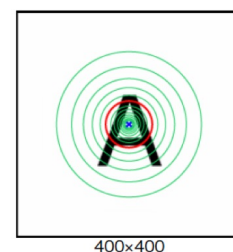
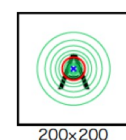
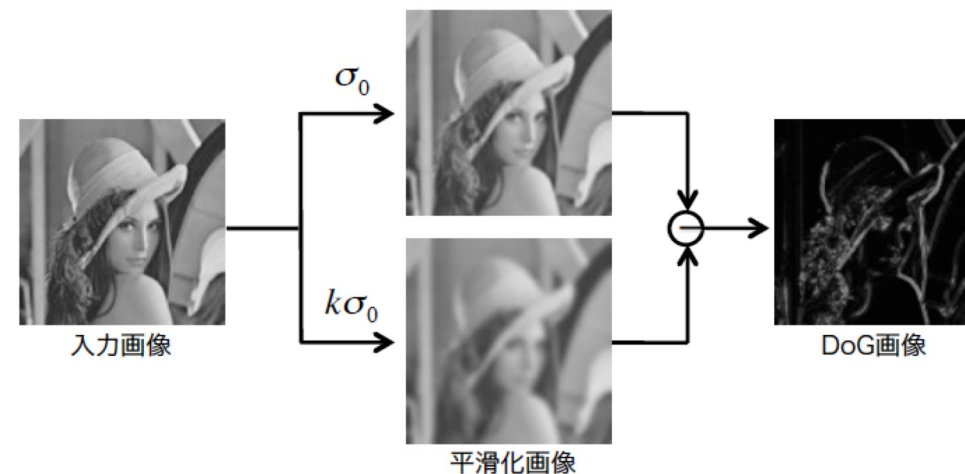
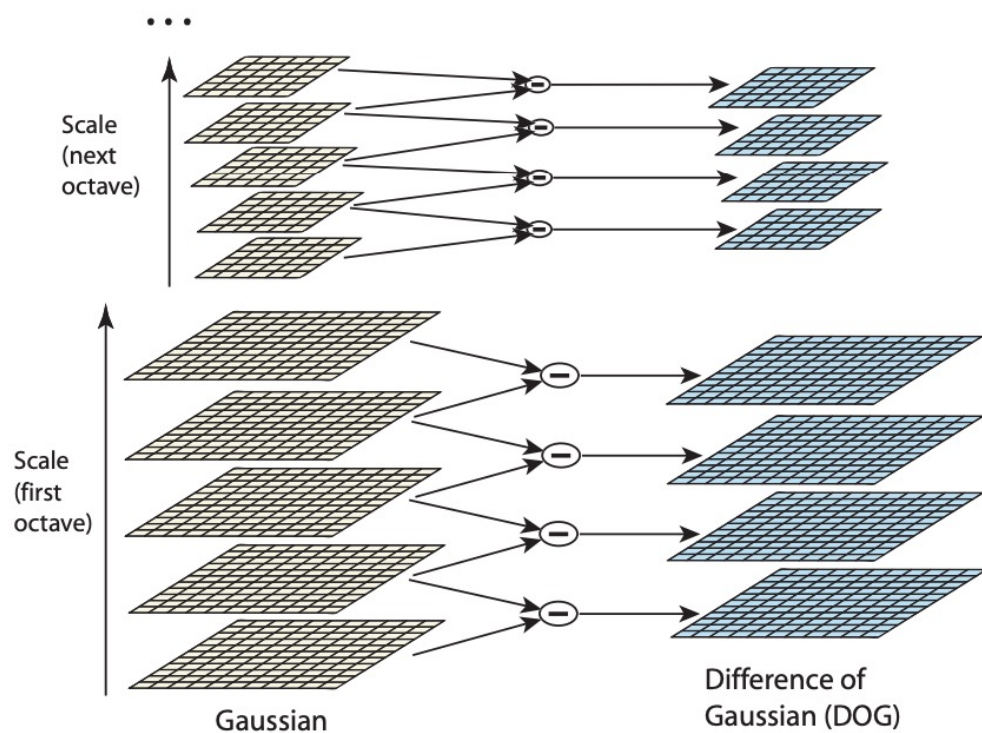
SIFT (2/4)

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Scale Invariant Feature

DoG Filter

◆ Fast Edge Extraction like LoG Filter



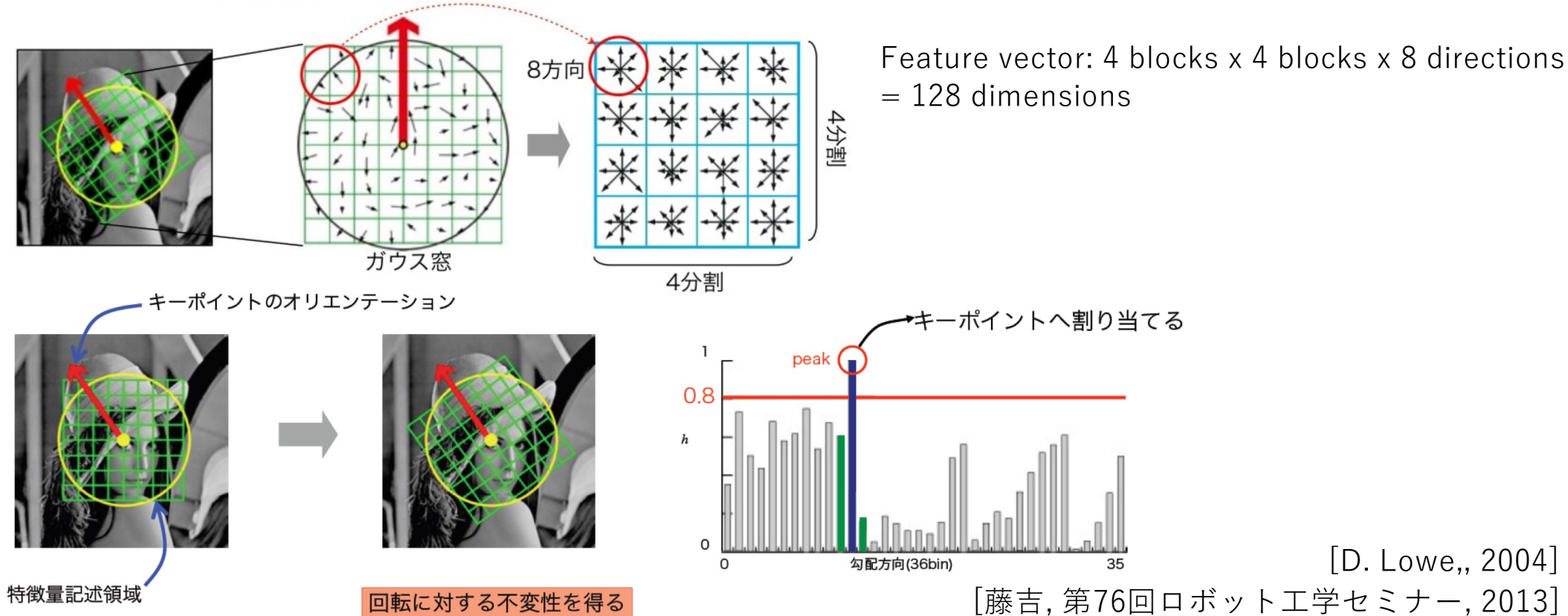
[D. Lowe., 2004]
[藤吉, 第76回ロボット工学セミナー, 2013]

SIFT (3/4)

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Rotation Invariant Feature

- Histogram of gradient direction around the key point



SIFT (4/4)

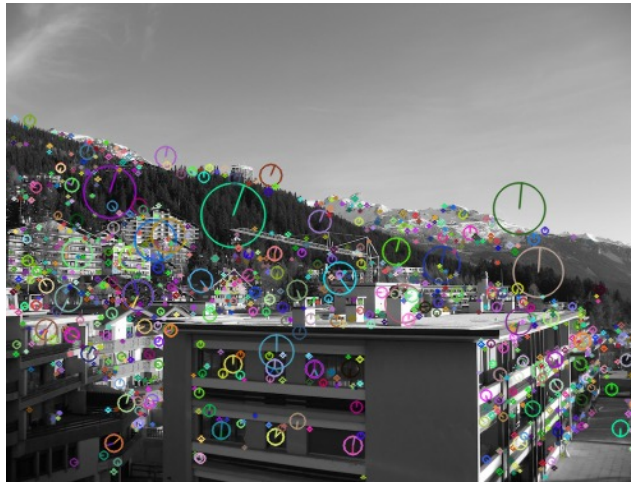
60

▣ SIFT: Scale-Invariant Feature Transform [D. Lowe, 1999, 2004]

- Rotation and Scale Invariant
- Robust for noise and Illumination change
- 128 feature vector for each key point
 - ◆ Normalized histogram of gradient direction



Original Image



SURF

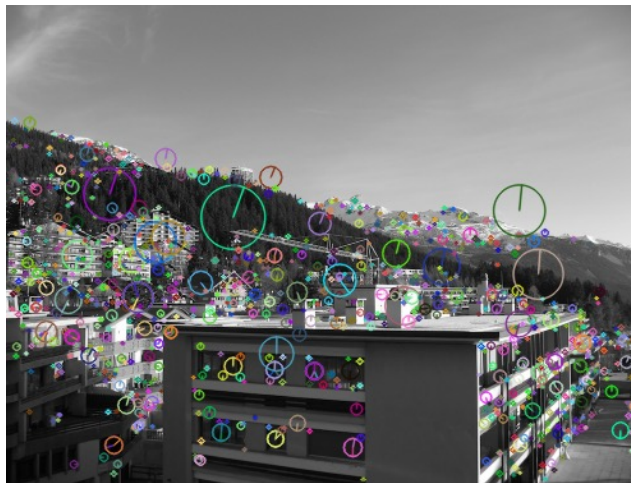
61

▣ SURF: Speeded Up Robust Feature [H. Bay, T. Tuytelaars, L. V. Gool, 2006]

- Fast rotation and scale invariant feature
- Almost 10 times faster than SIFT



Original Image



SIFT Key points



SURF Key points

Local Feature Matching

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- Find Corresponding key points for two images



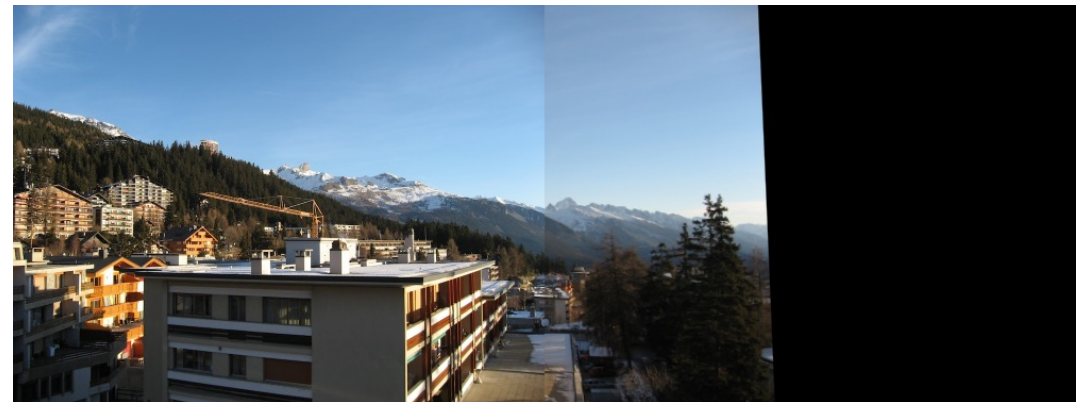
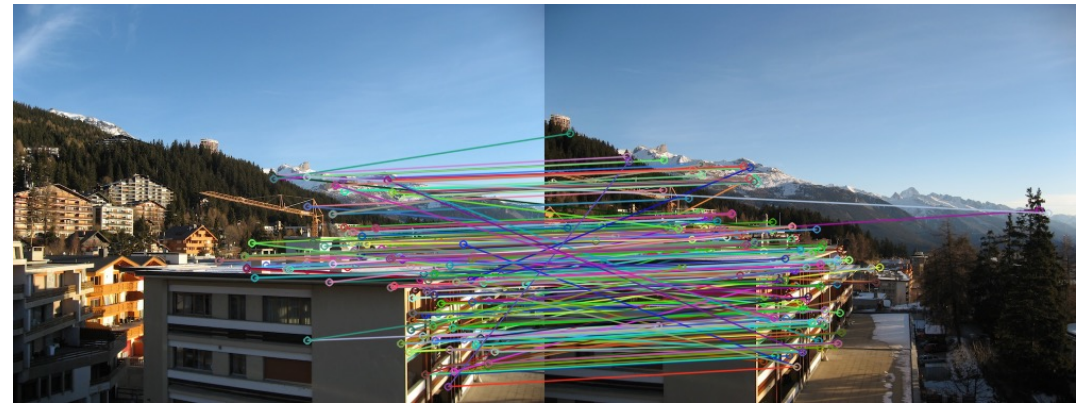
Image Stitching

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□ Feature matching

- Find Corresponding pixels
- Compute Homography Matrix

$$\blacklozenge \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = H \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$



Local Feature Matching

64

- Find Corresponding key points for two images



Application of Key Point Matching

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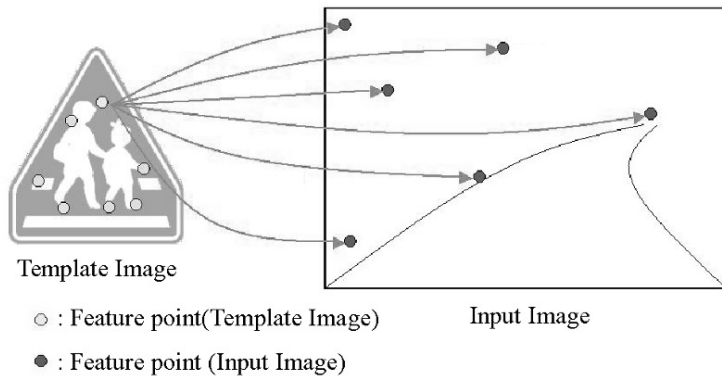


Fig. 2. Feature point matching.

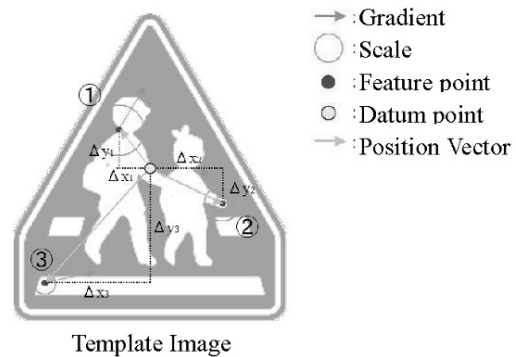


Fig. 3. Calculation of position vector.

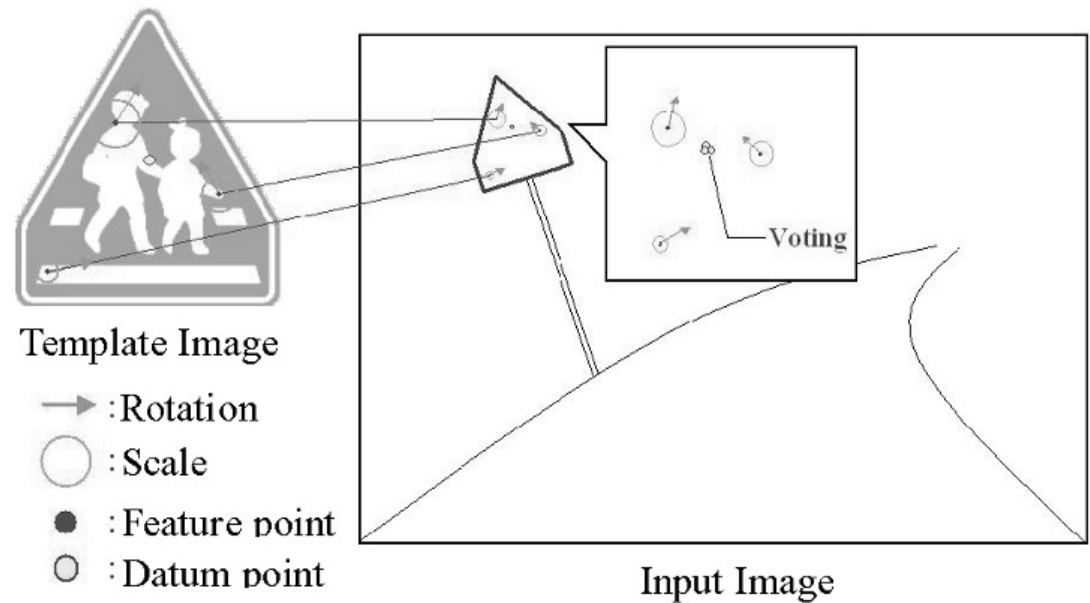


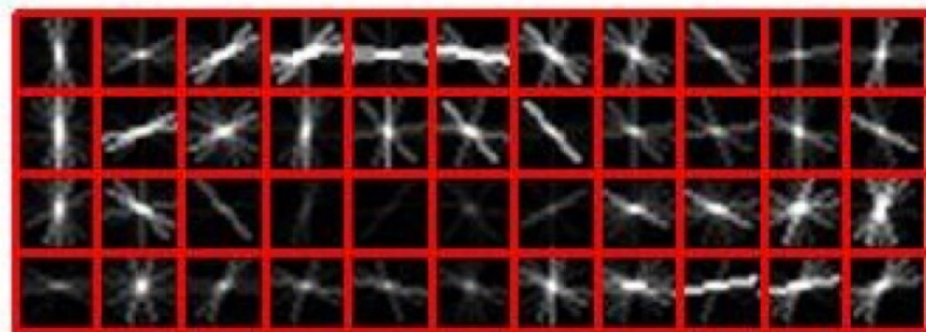
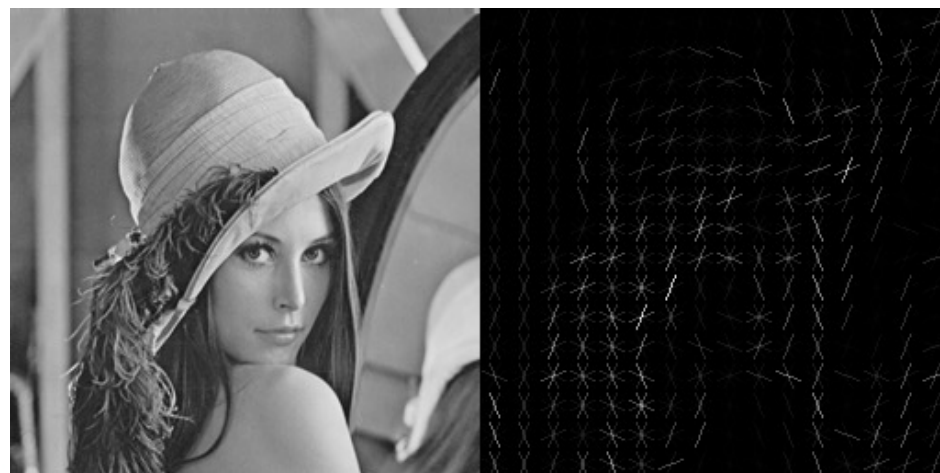
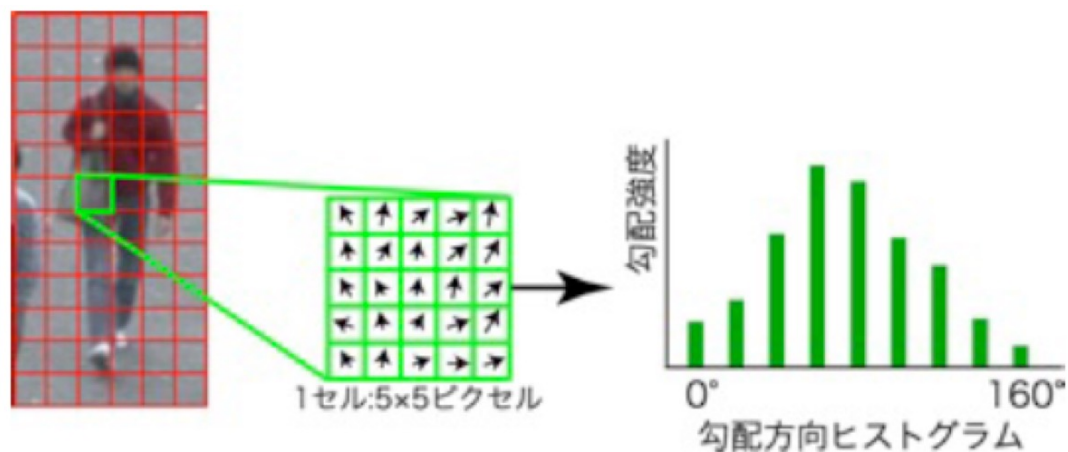
Fig. 5. Matching and Voting process of Feature points.

[高木, 藤吉, 2009]

HoG Feature

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□ Illumination Invariant Feature



Local Feature/Template Matching

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□ Local Feature

- Template Matching: Correlation
- Phase Only Correlation

○ Template Matching (1/5)

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□ Search template from the input image

- Raster Scan

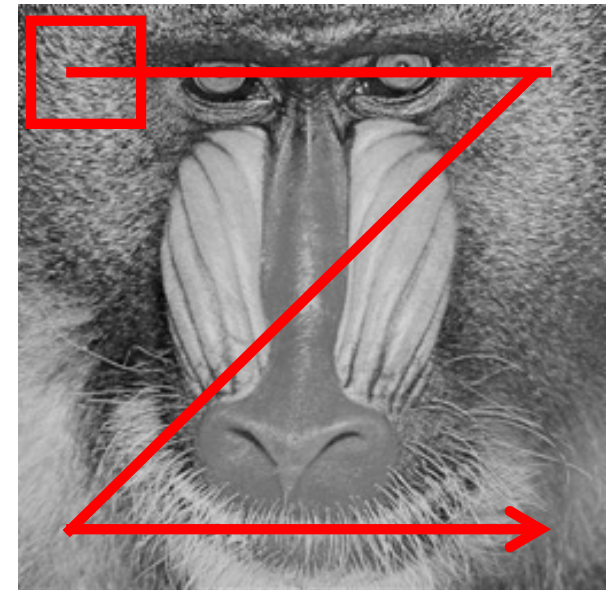
- ◆ Search the template image by shifting its position pixel by pixel
- ◆ Cost: image similarity



template

□ Image Similarity

- SSD: Sum of Squared Difference
- SAD: Sum of Absolute Difference
- NCC: Normalized Cross-Correlation
- ZNCC: Zero-mean Normalized Cross-Correlation



image

○ Template Matching (2/5)

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□ Search template from the input image

- Raster Scan

- ◆ Search the template image by shifting its position pixel by pixel
- ◆ Cost: image similarity

□ Image Similarity

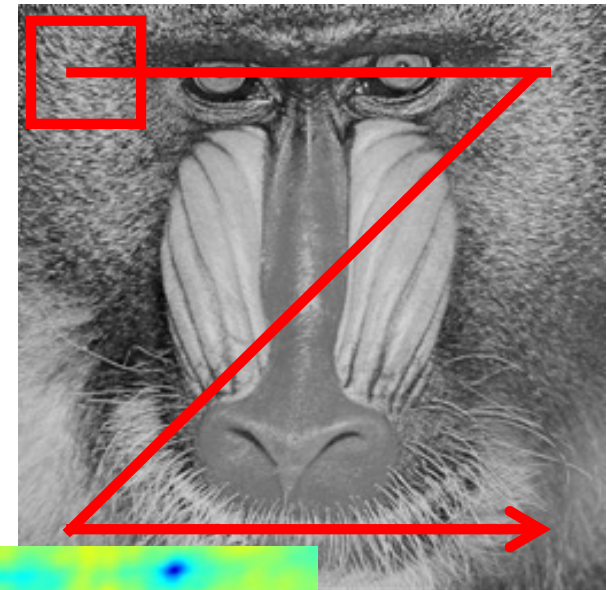
- SSD: Sum of Squared Difference

- ◆ $C_{SSD}(u, v) = \sum_{j=0}^{N-1} \sum_{i=0}^{M-1} (I(u + i, v + j) - T(i, j))^2$

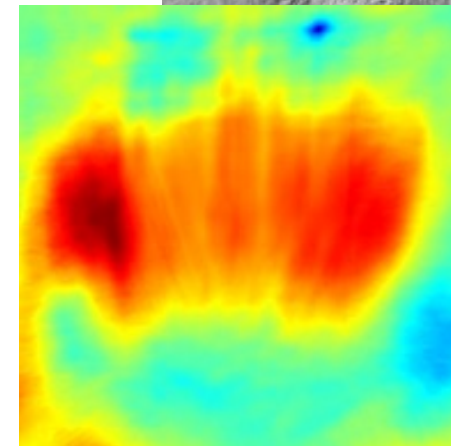
- $I(u, v)$: image
- $T(u, v)$: template image (image size: $M \times N$)
- (u, v) : pixel position



template



image



○ Template Matching (3/5)

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□ Search template from the input image

- Raster Scan

- ◆ Search the template image by shifting its position pixel by pixel
- ◆ Cost: image similarity

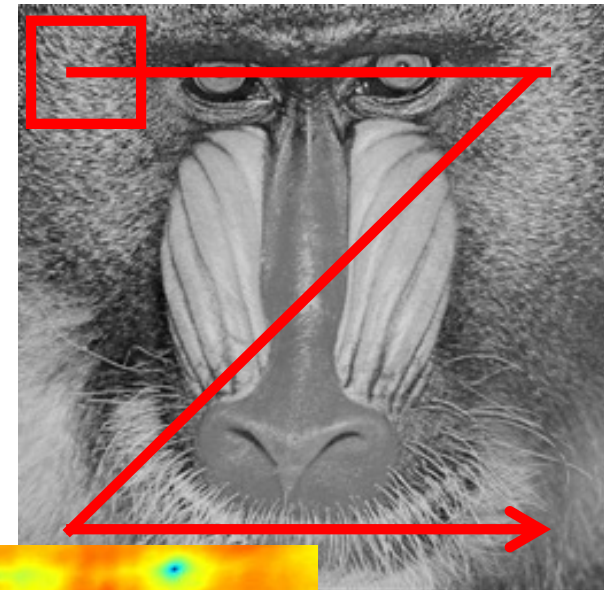
□ Image Similarity

- SAD: Sum of Absolute Difference

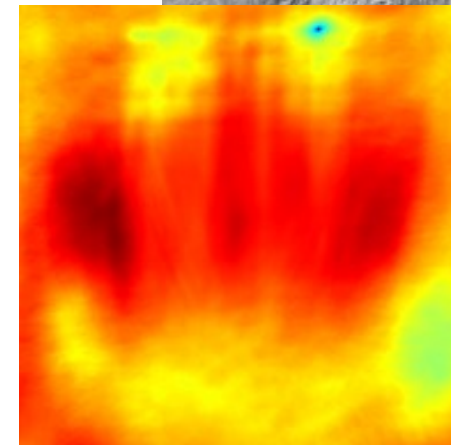
- ◆ $C_{SAD}(u, v) = \sum_{j=0}^{N-1} \sum_{i=0}^{M-1} |I(u + i, v + j) - T(i, j)|$
 - $I(u, v)$: image
 - $T(u, v)$: template image (image size: $M \times N$)
 - (u, v) : pixel position



template



image



○ Template Matching (4/5)

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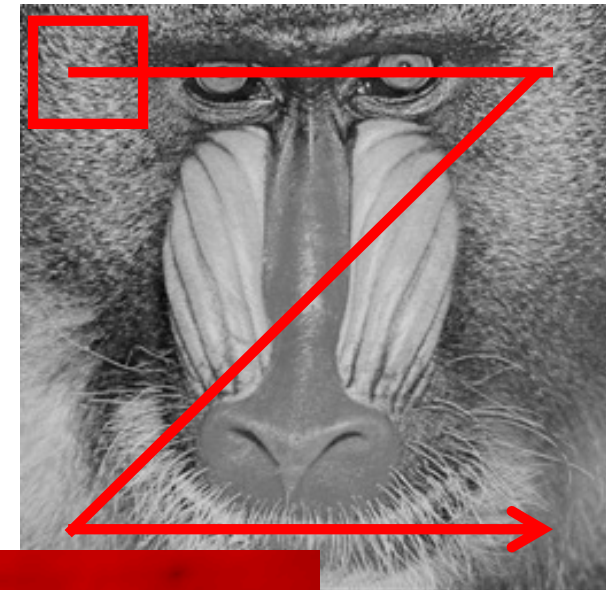
□ Search template from the input image

- Raster Scan

- ◆ Search the template image by shifting its position pixel by pixel
- ◆ Cost: image similarity



template



image

□ Image Similarity

- NCC: Normalized Cross-Correlation

$$\diamond C_{NCC}(u, v) = \frac{\sum_{j=0}^{N-1} \sum_{i=0}^{M-1} I(u+i, v+j) T(i, j)}{\sqrt{\sum_{j=0}^{N-1} \sum_{i=0}^{M-1} I(u+i, v+j)^2} \sqrt{\sum_{j=0}^{N-1} \sum_{i=0}^{M-1} T(i, j)^2}}$$

- $I(u, v)$: image
- $T(u, v)$: template image (image size: $M \times N$)
- (u, v) : pixel position
- Cosine similarity

$$\cos\theta = \frac{\mathbf{I} \cdot \mathbf{T}}{|\mathbf{I}| |\mathbf{T}|}$$



○ Template Matching (5/5)

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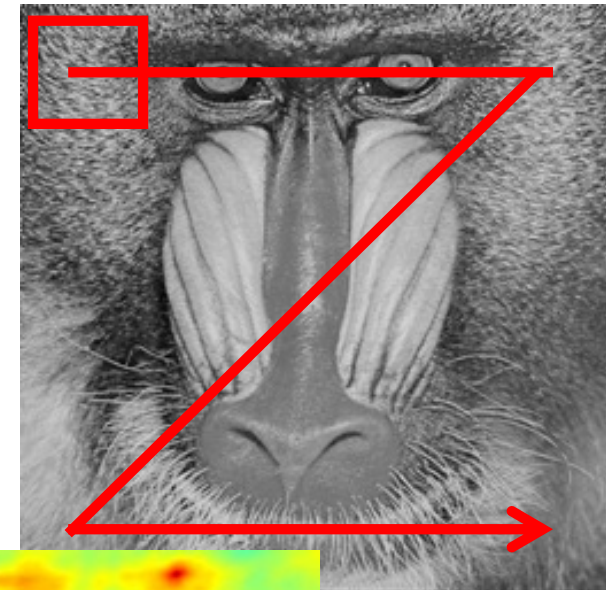
□ Search template from the input image

- Raster Scan

- ◆ Search the template image by shifting its position pixel by pixel
- ◆ Cost: image similarity



template



image

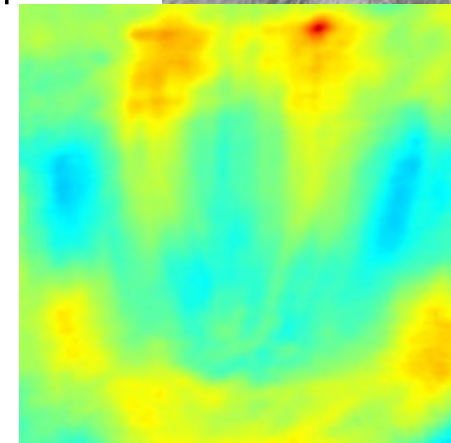
□ Image Similarity

- ZNCC: Zero-mean Normalized Cross-Correlation

$$\diamond C_{NCC}(u, v) = \frac{\sum_{j=0}^{N-1} \sum_{i=0}^{M-1} (I(u+i, v+j) - \bar{I})(T(i, j) - \bar{T})}{\sqrt{\sum_{j=0}^{N-1} \sum_{i=0}^{M-1} (I(u+i, v+j) - \bar{I})^2} \sqrt{\sum_{j=0}^{N-1} \sum_{i=0}^{M-1} (T(i, j) - \bar{T})^2}}$$

- $I(u, v)$: image
- $T(u, v)$: template image (image size: $M \times N$)
- (u, v) : pixel position
- $\bar{I}, \bar{T}$: average pixel value
- Cosine similarity

$$\cos\theta = \frac{\mathbf{I} \cdot \mathbf{T}}{|\mathbf{I}| |\mathbf{T}|}$$



Problem for Template Matching

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Image
(256, 256)



Template
(128, 128)

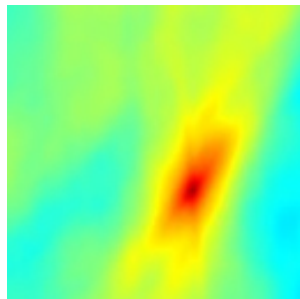
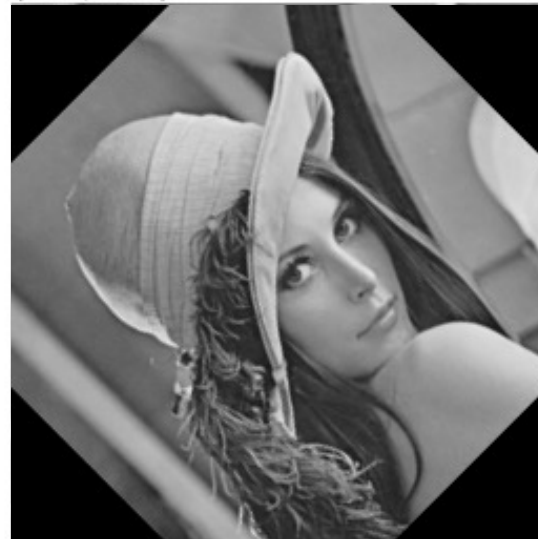
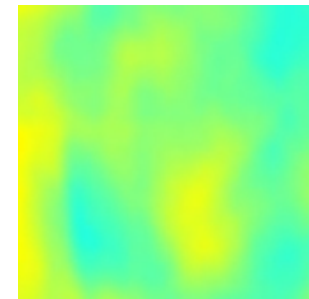


Image
(256, 256)



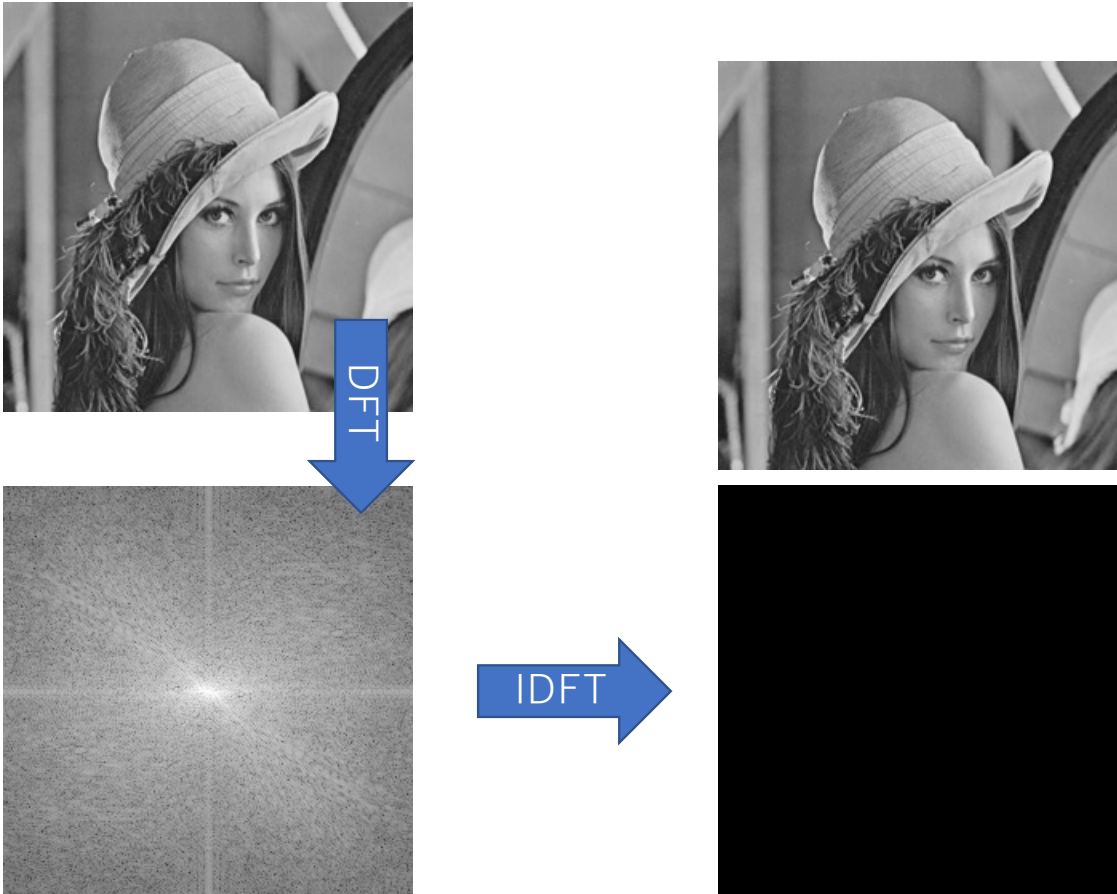
Template
(128, 128)



2D FFT

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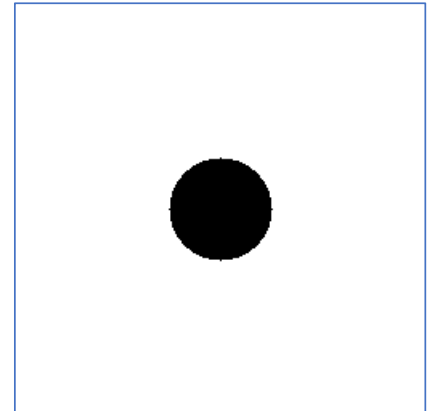
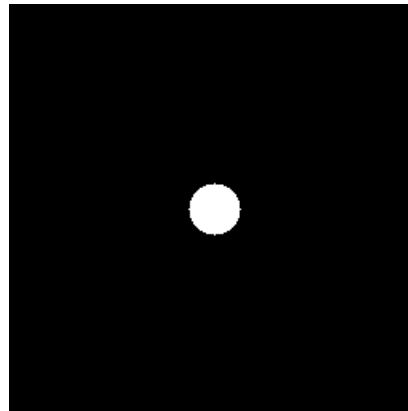
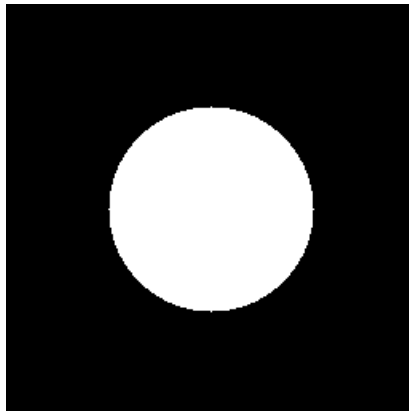
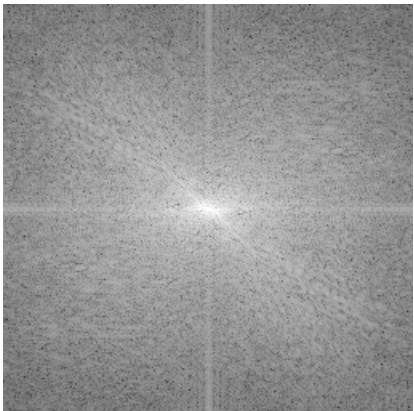
□ 2D FFT for image data



2D FFT

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□ Extract specific frequency



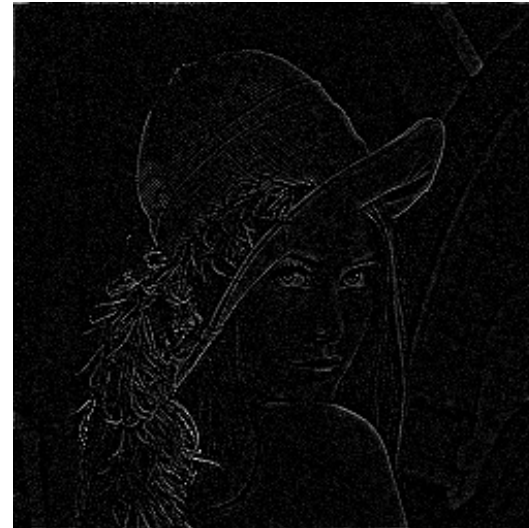
2D FFT

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□ Normalize based on Amplitude



Normalized Image
Max: 0.370
Min: -0.372



bias = 16



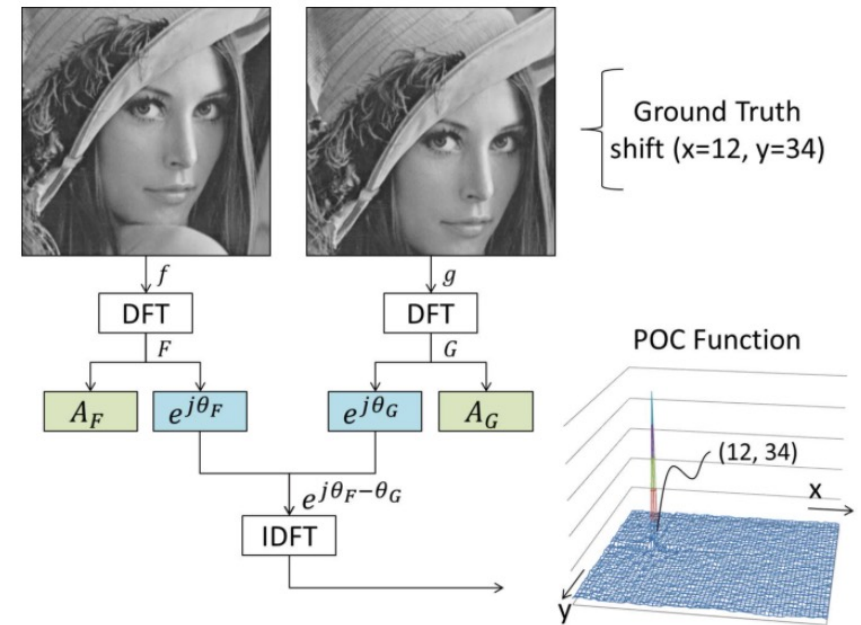
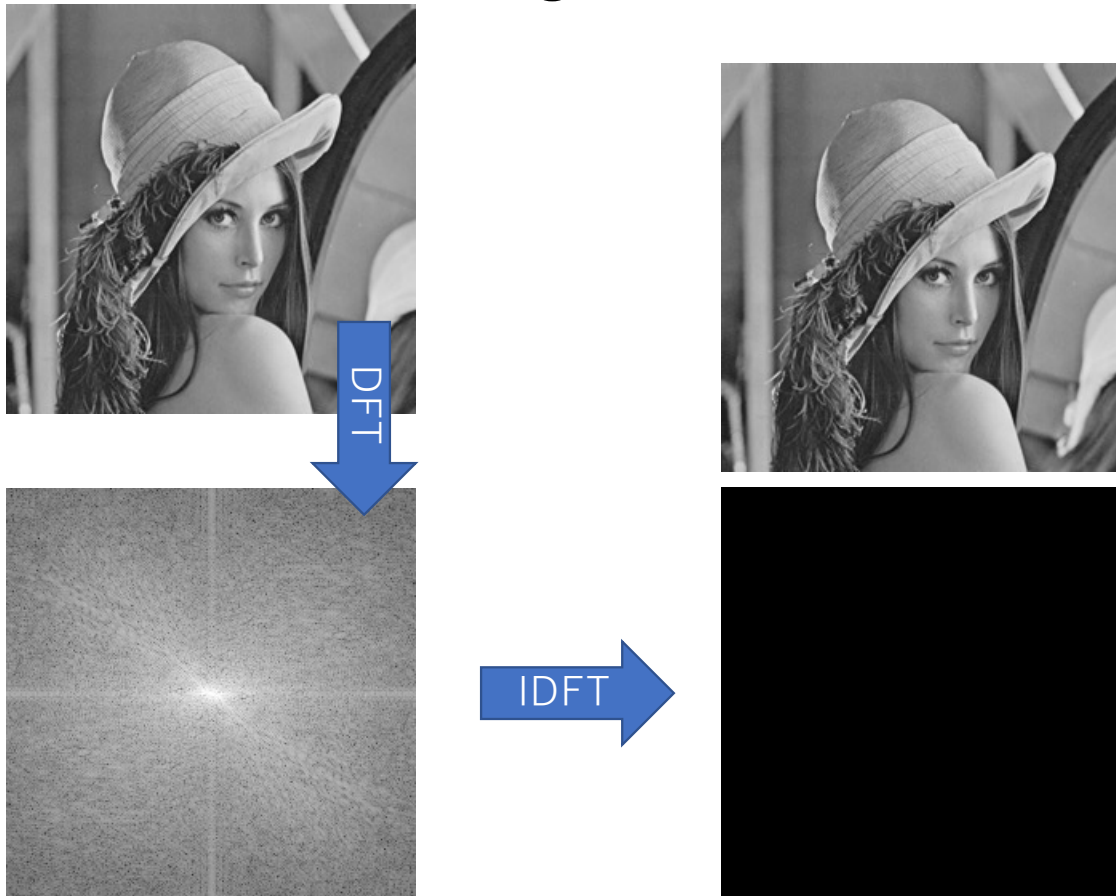
bias = 32

1. Extract greater than 0.0
 - If $val < 0.0$ then $val = 0.0$
2. Normalized Image
 - $val / maxVal * 255 + bias$
3. Clip Image from 0 to 255

Phase Only Correlation (1/2)

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2D FFT for image data



<http://www.sanko-shoko.net/note.php?id=13hc>

Phase Only Correlation (2/2)

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Image
(256, 256)

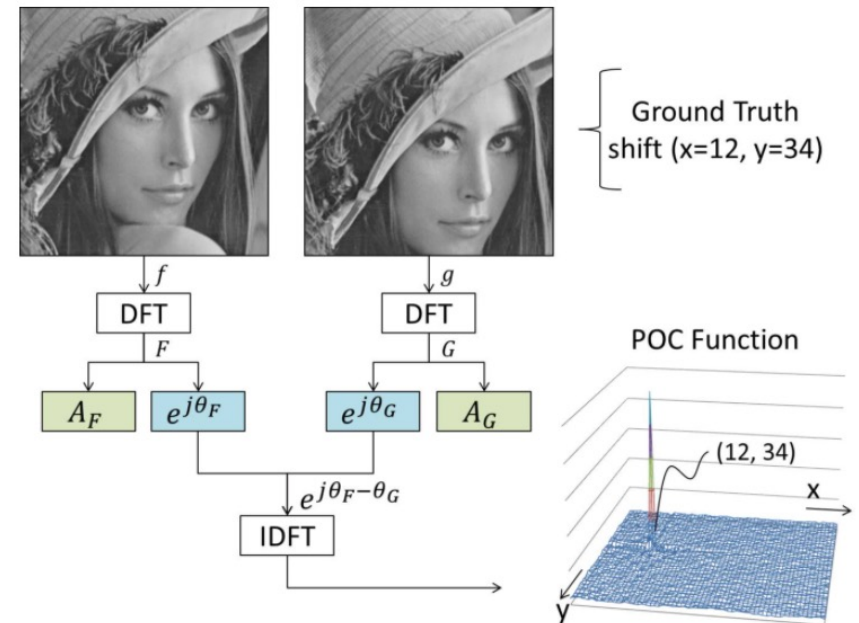


Template
(256, 256)



```
1 (x, y), res = cv2.phaseCorrelate(gray.astype(np.float32), template.astype(np.float32))  
2 print(x, y, res)
```

-79.93617845605024 -80.03547747384056 0.45598066941602156



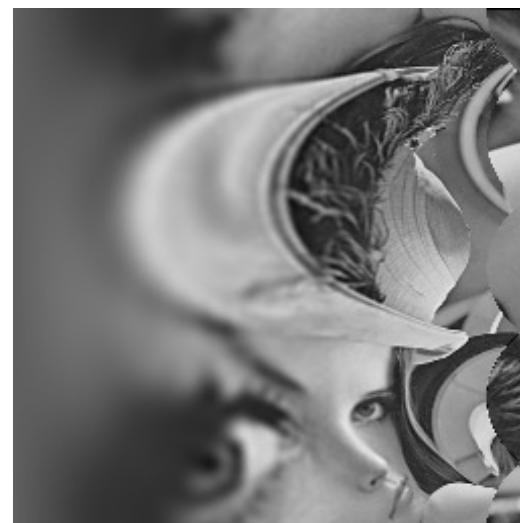
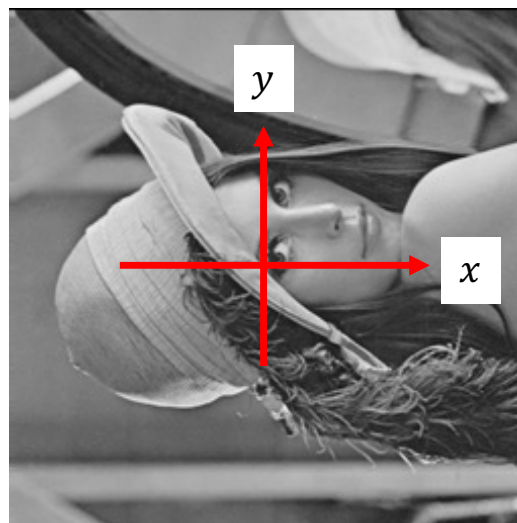
<http://www.sanko-shoko.net/note.php?id=13hc>

○ Log-Polar Image (1/2)

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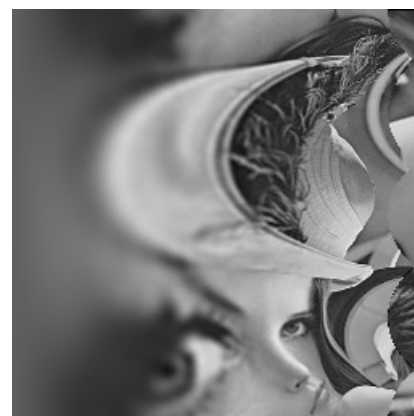
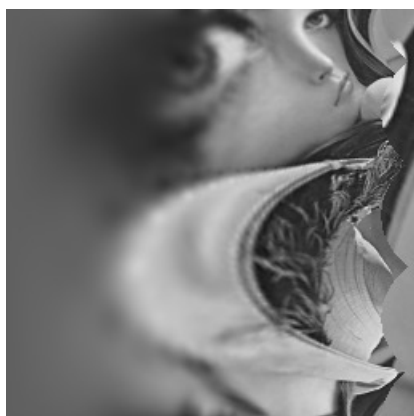
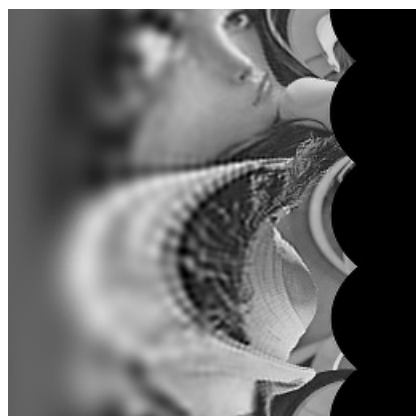
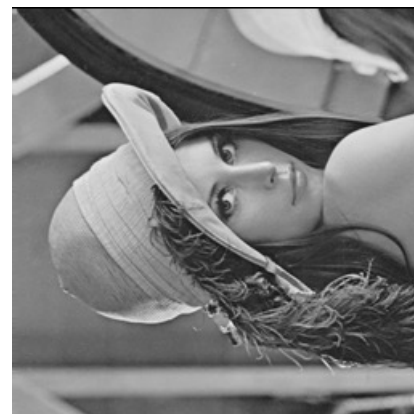
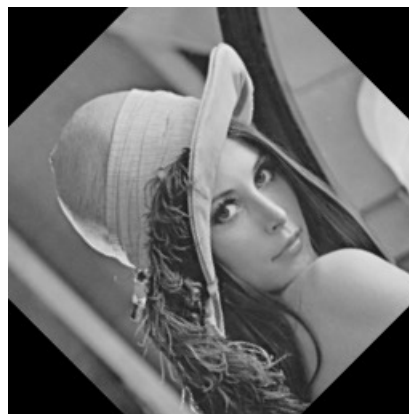
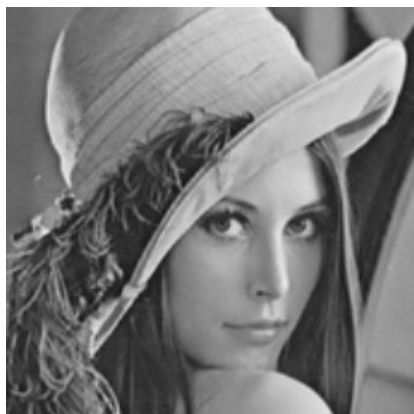
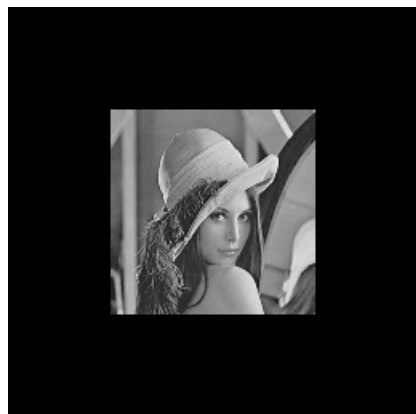
□ Convert Image to Polar Coordinate

- $$\begin{cases} \rho = \log \sqrt{x^2 + y^2} \\ \theta = \tan^{-1} y/x \end{cases}$$
- Rotation matching can be converted to 1D shifting



○ Log-Polar Image (2/2)

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Horizontal Shift

Horizontal Shift

Vertical Shift

Vertical Shift

Rotation Invariant POC (1/2)

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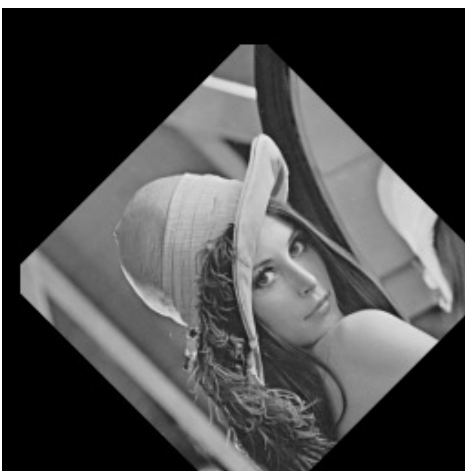


□ Application of POC

- Estimate Rotation, Scale, Position offset

□ Estimate Rotation, Scale

- Convert Log-Polar Image
 - ◆ Center position of transformation is important for Log-polar image



□ RIPOC

1. Apply FFT for both image
2. Convert Log-Polar Image
3. POC estimates rotation & scale offset
4. POC estimates position offset

Rotation Invariant POC (2/2)

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